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Machine And Method For Roll-Packing Of Mattresses

Abstract

A roll-packing machine (**100**) is provided with a belt drive mechanism having a closed-loop belt (**110**). Further, the roll-packing machine (**100**) is provided with a feed mechanism (**130, 140**) configured to feed a mattress (**10**) to the belt (**110**). The belt drive mechanism is configured to form an inner loop (**115**) of the belt, to receive the mattress (**10**) in the inner loop (**115**), and to roll the mattress (**10**) by advancing the belt (**110**) while the mattress (**10**) is received in the inner loop (**115**). The inner loop (**115**) of the belt may thus act as a rolling chamber for the mattress (**10**).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 17/606,295 filed Oct. 25, 2021 (Pending), which claims priority to International Patent Application No. PCT/EP2020/062310 filed May 4, 2020 (pending), the disclosures of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates to a roll-packing machine for packaging of mattresses and to a method of roll-packing a mattress.

BACKGROUND OF THE INVENTION

[0003] With respect to packaging mattresses, a known packaging technique is roll packing. In this case the finished mattress is rolled to form a compact roll. Bands, tape or an outer wrapping may then be used to hold the roll in its form during storage or shipment. The roll may be formed manually. Alternatively, the roll may be formed in an automated manner, using a roll-packing machine. As described in WO 2018/049812 A1, such roll-packing machine may use a substantially cylindrical rolling chamber formed of a plurality of rollers.

[0004] However, using a rolling chamber formed of a plurality of rollers may result in rather high complexity of the roll-packing machine, e.g., because the arrangement of the rollers needs to be adjustable in order to conform to the outer shape of the mattress roll being formed and to allow removal of the mattress roll from the rolling chamber. Further, there may also be a risk of the mattress getting jammed between the rollers.

[0005] Accordingly, there is a need for techniques which allow for efficiently for roll-packing mattresses.

BRIEF SUMMARY OF THE INVENTION

[0006] The present invention provides a roll-packing machine according to claim **1** and to a method according to claim **12**. The dependent claims define further embodiments.

[0007] Accordingly, an embodiment of the invention provides a roll-packing machine for packaging a mattress. The roll-packing machine comprises a belt drive mechanism having a closed-loop belt. Further, the roll-packing machine comprises a feed mechanism configured to feed the mattress to the belt. The belt drive mechanism is configured to form an inner loop of the belt, to receive the mattress in the inner loop, and to roll the mattress by advancing the belt while the mattress is received in the inner loop. The inner loop of the belt may thus act as a rolling chamber for the mattress.

[0008] According to an embodiment, the belt may be a segmented belt, i.e., be formed of multiple segments joined to each other by hinge joints. Flexibility of the belt may thus be provided by the pivot joints, rather than by the material of the segments. In this way, the belt may be provided with sufficient intrinsic stability to maintain the inner loop, without requiring dedicated support structures for maintaining the shape of the inner loop. Rather, stability of the inner loop may be provided by a first and a second roller, between which the inner loop is formed, by tension of the belt, and by a force generated by the mattress which is being fed to a portion of the belt between the first roller and the second roller.

[0009] According to an embodiment, the belt drive mechanism is configured to adjust a size of the inner loop. In this way, different sizes of the finished mattress roll may be accommodated. Further, the size of the inner loop may be controlled in accordance with the diameter of the mattress roll while the mattress is being rolled. In particular, the belt drive mechanism may be configured to increase the size of the inner loop while the mattress is being rolled.

[0010] According to an embodiment, the belt drive mechanism comprises a first roller and a second roller, with the inner loop being formed between the first roller and the second roller. In this case, the belt drive mechanism may be configured to adjust the size of the inner loop by individually controlling rotation of the first roller and the second roller. For example, with respect to an advancement direction of the belt, the first roller may be arranged upstream of the portion forming the inner loop, and the second roller may be arranged downstream of the portion forming the inner loop, i.e., the belt may be advanced from the first roller towards the second roller. In this case, the size of the inner loop may be increased by rotating the first roller at a higher rate than the second roller. The different rates of rotation may be achieved by providing the first roller and the second roller with individually controllable drive mechanisms. In addition or as an alternative, the first roller and the second roller could be provided with individually controllable brake mechanisms. By individually controlling rotation of the first roller and the second roller, the size of the inner loop may be adjusted in a highly efficient manner, without requiring complex mechanical structures.

[0011] According to an embodiment, the roll packing machine further comprises a belt tensioning mechanism configured to control tension of the belt while adjusting the size of the inner loop. In this way, changes of the size of the inner loop may be compensated and the tension of the belt may be kept at a desired level.

[0012] According to an embodiment, the tensioning mechanism is configured to control the tension of the belt by adjusting a size of one or more additional loops of the belt. For example, the belt drive mechanism may comprise multiple rollers for guiding the belt. At least some of the rollers may form the one or more additional loops. The tensioning mechanism may then be configured to adjust the size of the one or more additional loops by displacing two or more of the rollers relative to each other. By using a higher number of the additional loops, a required displacement range may be reduced, which may be beneficial in view of space-requirements for the roll-packing machine.

[0013] According to an embodiment, the belt drive mechanism comprises a first roller and a second roller, with the inner loop being formed between the first roller and the second roller. In this case, the roll-packing machine may further comprises an ejection mechanism configured to eject the rolled mattress from the inner loop by displacing the first roller away from the second roller. Due to the displacement, the inner loop may be opened so as to enable release and removal of the finished mattress roll from the inner loop. In some cases, the displacement may even have the effect that the portion of the belt between the first roller and the second roller takes a substantially straight-line form, so that the finished mattress roll can be easily moved from the roll-packing machine by rolling along the belt and/or by advancing the belt.

[0014] According to an embodiment, the roll-packing machine is further configured to wrap the rolled mattress by feeding a wrapping material, e.g., a plastic film, into the inner loop with the rolled mattress and advancing the belt to rotate the rolled mattress in the inner loop and wrap the wrapping material around the rolled mattress. In this case, the advancement of the belt may pull the wrapping material into the space between the rolled mattress and the inner loop. As a result, wrapping of the rolled mattress and formation of the finished mattress roll may be achieved in a highly efficient manner and with only a limited number of additional components, e.g., for feeding the wrapping material into the loop and/or for welding the wrapping material.

[0015] According to an embodiment, the feed mechanism is configured to compress the mattress while feeding the mattress to the belt. In this way, formation of a more compact mattress roll can be achieved.

[0016] According to a further embodiment, a method of packaging a mattress is provided. The method may be performed by the above roll-packing machine. The method comprises feeding the mattress to a closed-loop belt of a belt drive mechanism, forming an inner loop of the belt,

receiving the mattress in the inner loop, and rolling the mattress by advancing the belt while the mattress is received in the inner loop. As mentioned above, the belt may be a segmented belt.
[0017] According to an embodiment, the method further comprises adjusting the size of the inner loop while the mattress is being rolled, e.g., by increasing the size of the inner loop so as to accommodate an increasing diameter of the mattress roll being formed.
[0018] According to an embodiment, the method further comprises wrapping the rolled mattress by feeding a wrapping material, e.g., a plastic film, into the inner loop with the rolled mattress and advancing the belt to rotate the rolled mattress in the inner loop and wrap the wrapping material around the rolled mattress. In this case, the advancement of the belt may pull the wrapping material into the space between the rolled mattress and the inner loop. As a result, wrapping of the rolled mattress and formation of the finished mattress roll may be achieved in a highly efficient manner.
[0019] According to an embodiment, the belt drive mechanism comprises a first roller and a second roller, with the inner loop being formed between the first roller and the second roller. In this case, the method may further comprise ejecting the rolled mattress from the inner loop by displacing the first roller away from the second roller. Due to the displacement, the inner loop may be opened so as to enable release and removal of the finished mattress roll from the inner loop. In some cases, the displacement may have the effect that the portion of the belt between the first roller and the second roller takes a substantially straight-line form, so that the finished mattress roll can be moved from the roll-packing machine by rolling along the belt and/or by advancing the belt.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0020] Embodiments of the invention will be described with reference to the accompanying drawings.
[0021] FIG. **1**A illustrates a mattress to be roll-packed according to an embodiment.
[0022] FIG. **1**B illustrates a mattress roll as produced according to an embodiment.
[0023] FIG. **2** schematically illustrates a sectional view of a roll-packing machine according to an embodiment.
[0024] FIG. **3** schematically illustrates a sectional view of a roll-packing machine according to a further embodiment.
[0025] FIGS. **4**A and **4**B schematically illustrates elements of a segmented belt as utilized in a roll-packing machine according to an embodiment.
[0026] FIGS. **5**A-**5**H schematically illustrate different stages of a roll-packing process according to an embodiment.
[0027] FIG. **6** shows a flowchart for illustrating a method according to an embodiment of the invention.
DETAILED DESCRIPTION OF EMBODIMENTS
[0028] Exemplary embodiments of the invention will be described with reference to the drawings. In particular, a roll-packing machine and a roll-packing method for mattresses will be described. While the following detailed description refers to packaging of mattresses as typically used for bedding furniture, it is to be understood that the illustrated concepts are not limited to this type of mattress, but for example could also be applied to other types mattress, e.g., like used in other types of furniture or even in applications not related to furniture, e.g., mattresses as used for sports or as components of buildings or vehicles. The mattresses may for example be based on foam material, one or more innerspring units, e.g., based on pocketed springs or a combination of pocketed springs and wire framework elements, or on a combination of a foam material and one or more innerspring units. The roll-packing machine and the method of the illustrated concepts are used for packaging the mattress by rolling, thereby forming a mattress roll which includes the rolled mattress and

optionally also and outer wrapping around the rolled mattress. In some cases, the mattress may also be wrapped before being rolled, i.e., which means that the mattress roll may also include an inner wrapping formed around the mattress itself. It is noted that the features of different embodiments may be combined with each other unless specifically stated otherwise.

[0029] FIG. **1**A shows a sectional view for schematically illustrating a mattress **10**. As mentioned above, the mattress **10** may be a mattress for bedding applications. The mattress may for example be based on foam material and/or on one or more innerspring units. If present, such innerspring unit may be based on pocketed springs or a combination of pocketed springs and wire framework elements. The mattress may have a substantially box-shaped outer form, with a typical length being in the range of 100 cm to 250 cm, a typical width being in the range of 60 cm to 200 cm, and a typical thickness being in the range of 5 cm to 30 cm. The mattress **10** is assumed to be compressible in the thickness direction.

[0030] FIG. **1**B illustrates a mattress roll **20** which includes the rolled mattress **10** and an outer wrapping **25** enclosing the rolled mattress **10**. The outer wrapping may for example be formed of one or more layers of a plastic film. The outer wrapping **25** ensures that the mattress **10** stays in the rolled state and thus keeps the mattress roll stable, e.g., during storage or transport. A typical outer diameter of the mattress roll is in the range of 30 cm to 80 cm, more specifically in the range of 40 cm to 60 cm. For forming the mattress roll **20**, the mattress **10** is typically rolled around an axis parallel to its width direction. In some cases, an additional wrapping or other type of cover layer may also be provided on the mattress **10** before being rolled, resulting in the formation of inner layers of wrapping or cover material also in between the windings of the mattress roll **20**.

[0031] FIG. **2** shows a sectional view for schematically illustrating a roll-packing machine **100** which may be used for automated roll-packing of mattresses. In the following explanations, it is assumed that the roll-packing machine **100** is used for roll packing the mattress **10**, thereby producing the mattress roll **20**. In FIG. **2**, a first horizontal direction is denoted by “x”, and a vertical direction is denoted by “z”. A second horizontal direction, extending perpendicular to the x-direction and the y-direction, i.e., perpendicular to the drawing plane, is denoted by “y”. In the illustrated example, the roll-packing machine **100** is configured to perform a roll-packing process involving rolling of the mattress **10** around an axis in the y-direction.

[0032] As illustrated, the roll-packing machine **100** is provided with a belt-drive mechanism with a closed-loop belt **110** which is guided by rollers **121**, **122**, **123**, **124**, **125**, **126**. Rotational axes of the rollers **121**, **122**, **123**, **124**, **125**, **126** extend in the y-direction. In the illustrated example, the belt **110** is assumed to be driven by the rollers **121** and **122**. For this purpose, the rollers **121**, **122** are provided with individually controllable drive mechanisms, e.g., respectively based on an electric motor coupled to a rotation shaft of the roller **121**, **122**. An advancement direction of the belt **110** is illustrated by arrows A.

[0033] As further illustrated, the belt drive mechanism is configured to form an inner loop **115** of the belt **110**. As will be further explained below, the inner loop **115** acts as a rolling chamber for rolling the mattress **10**. The inner loop **115** is formed in a portion of the belt **110** between the roller **121** and the roller **122**. A size of the loop **115**, i.e., the length of the portion of the belt **110** between the roller **121** and the roller **122**, can be adjusted by individually controlling rotation of the roller **121** and the roller **122**. When for example assuming an advancement of the belt **110** in the direction of the arrows A, the size of the loop **115** can be increased by controlling the roller **121** to rotate faster than the roller **122**. Similarly, the size of the loop **115** can be decreased by controlling the roller **121** to rotate slower than the roller **122**. Since the rollers **121**, **122** are located at an entry into the inner loop **115**, the rollers **121**, **122** may also be referred to as entry rollers.

[0034] In the illustrated example, the roll-packing machine **100** is further provided with a tensioning mechanism for controlling tension of the belt **110**. By means of the tensioning mechanism, the tension of the belt **110** can be maintained at a desired level, even when changing the size of the inner loop **115**. In the example of FIG. **2**, the tensioning mechanism includes a drive mechanism **150** for displacing the roller **124** of the belt drive mechanism with respect to other rollers **123** of the belt drive mechanism. The displacement of the roller **124** is illustrated by arrow B. By the displacement of the roller **124**, a size of an additional loop of the belt **110**, which is formed by the rollers **123**, **124**, and **125**, can be adjusted. Specifically, the size of the additional loop can be set in such a way that it compensates changes of the size of the inner loop **115**. For example, while increasing the size of the inner loop **115**, the required additional length of the belt portion forming the inner loop **115** can be provided by reducing the size of the additional loop formed by the rollers **123**, **124**, and **125**.

[0035] In the illustrated example, the roll-packing machine **100** is further provided with an ejection mechanism for allowing release and removal of the finished mattress roll **20** from the inner loop **115**. In the illustrated example, the ejection mechanism is based on a drive mechanism which displaces the entry roller **121** away from the entry roller **122**. In the example of FIG. **2**, the rollers **121** and **126** of the belt drive mechanism are mounted on a tiltable support structure **155**. By tilting the support structure as illustrated by arrow C, the entry roller **121** can be moved away from the entry roller **122**, resulting in the inner loop **115** being opened, so that the mattress roll **20** formed in the inner loop **115** is released and can be removed.

[0036] In the illustrated example, the roll-packing machine **100** is further provided with a wrapping mechanism **160** for wrapping the rolled mattress **10** with a wrapping material, thereby forming the outer wrapping **25**. In the illustrated example, the wrapping mechanism **160** is configured to introduce the wrapping material **25**, which is supplied from a supply roll **161**, into the inner loop **115** with the rolled mattress **10**. A tool **162** may then be used for cutting and/or welding the wrapping material wrapped around the rolled mattress **10**.

[0037] The roll-packing machine **100** further includes a feed mechanism for feeding the mattress **10** to the belt **110**, in particular to the portion of the belt **110** between the entry rollers **121**, **122**. In the illustrated example, the feed mechanism includes a lower conveyor belt mechanism **130** and an upper conveyor belt mechanism **140**. The lower conveyor belt mechanism **130** includes a belt **131** guided by rollers **132**, **133**, **134**. One of the rollers **132**, **133**, **134**, e.g., the roller **132**, may be used for driving the belt **131**. The upper conveyor belt mechanism **140** includes a belt **141** guided by rollers **142**, **143**. One of the rollers **142**, **143**, e.g., the roller **143**, may be used for driving the belt **131**. The mattress **10** is conveyed by advancing the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140** in a direction as illustrated by arrow D. The feed mechanism may be opened for receiving the mattress. In the illustrated example, this is achieved by a drive mechanism **145** which displaces the roller **142** in a direction as illustrated by arrow D. By means of the drive mechanism **145** a part of the upper conveyor belt mechanism **140** may be moved away from the lower conveyor belt mechanism drive **130** to facilitate insertion of the mattress in an initial stage of the roll-packing process. After inserting the mattress **10**, the drive mechanism **145** may be used to move the upper conveyor belt mechanism **140** toward the lower conveyor belt mechanism **130** so that the mattress **10** is sandwiched between the belt **131** of the lower conveyor belt mechanism **130** and the belt **141** of the upper conveyor belt mechanism **140**. In some scenarios, this may also result in compression of the mattress **10** between the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140**. The distance between the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140** may be adjustable to accommodate different mattress thicknesses and/or to adjust a degree of compression of the mattress **10**.

[0038] FIG. **3** shows a sectional view for schematically illustrating a further roll-packing machine **100′** which may be used for automated roll-packing of mattresses, e.g., for roll packing the mattress **10**, thereby producing the mattress roll **20**. Similar to FIG. **2**, in FIG. **3** a first horizontal direction is denoted by “x”, and a vertical direction is denoted by “z”. A second horizontal direction, extending perpendicular to the x-direction and the y-direction, i.e., perpendicular to the drawing plane, is denoted by “y”. In the illustrated example, the roll-packing machine **100′** is configured to perform a

roll-packing process involving rolling of the mattress **10** around an axis in the y-direction.
[0039] In many aspects, the roll-packing machine **100**′ of FIG. **3** is similar to that of FIG. **2**, and corresponding components of the roll-packing machine **100** and the roll-packing machine **100**′ are denoted by the same reference numerals. Further details concerning structure and functionality of these components in the roll-packing machine **100**′ can be taken from the description in connection with FIG. **2**.
[0040] The roll-packing machine **100**′ differs from the roll-packing machine **100** in that the belt drive mechanism includes more additional loops for controlling the tension of the belt **110**. In the roll-packing machine **100**, the belt-drive mechanism includes rollers **121**, **122**, **123**A, **123**B, **123**C, **124**, **125**, and **126** for guiding the belt **110**. The rollers **121**, **122**, **124**, **125**, and **126** are similar to the rollers **121**, **122**, **124**, **125**, and **126** in the roll-packing machine **100**. The rollers **123**A, **123**B, **123**C replace the roller **123** of the roll-packing machine **100**, thereby forming two additional loops (two outer loop, one inner loop) of the belt **100**.
[0041] In the roll-packing machine **100**′ the tensioning mechanism for controlling tension of the belt **110** includes a drive mechanism **150**′ for displacing the rollers **123**B and **124** of the belt drive mechanism with respect to other rollers of the belt drive mechanism. The displacement of the roller **123**B, **124** is illustrated by arrows B. By the displacement of the rollers **123**B, **124**, the sizes of the additional loop of the belt **110**, which are formed by the rollers **123**A, **123**B, **123**C, **124**, and **125**, can be adjusted. Specifically, the sizes of the additional loops can be set in such a way that they compensate changes of the size of the inner loop **115**. For example, while increasing the size of the inner loop **115**, the required additional length of the belt portion forming the inner loop **115** can be provided by reducing the sizes of the additional loops formed by the rollers **123**A, **123**B, **123**C, **124**, and **125**. In the roll-packing machine **100**′, the required range of displacement of the rollers **123**B, **124** is lower than the required range of displacement of the roller **124** in the roll-packing machine **100**. This may for example allow for a reduced space requirement of the roll-packing machine **100**′, e.g., in terms of a dimension of the roll-packing machine **100**′ in the x-direction.
[0042] FIGS. **4**A and **4**B further illustrate an implementation of the belt **110** which may be used in the roll-packing machine **100** or the roll-packing machine **100**′. More specifically, FIG. **4**A shows a top view of a part of the belt **110**, and FIG. **4**B shows a side view of a part of the belt **110**. In FIGS. **4**A and **4**B, a length direction of the belt **110** (along which the belt **110** is advanced in operation of the roll-packing machine **100**, **100**′) is denoted by L, a width direction of the belt **110** (parallel to the axis of rolling the mattress **10**) is denoted by W, and a thickness direction of the belt **110** is denoted by T.
[0043] In the illustrated example, the belt **110** is implemented as a segmented belt, i.e., formed of multiple segments **111**, **112**, **113**, **114** which are joined together by hinged couplings. The segments **111**, **112**, **113**, **114** themselves are assumed to be relatively rigid, with deformation of the segments **111**, **112**, **113**, **114** during operation of the roll-packing machine **100**, **100**′ being negligible. Flexibility of the belt **110** may thus be provided by the hinged joints couplings between the segments **111**, **112**, **113**, **114**, rather than by the material of the segments **111**, **112**, **113**, **114**. In this way, the belt may be provided with sufficient intrinsic stability to maintain the inner loop **115**, without requiring dedicated support structures for maintaining the shape of the inner loop **115**. Rather, stability of the inner loop may be provided by the entry rollers **121**, **122**, between which the inner loop **115** is formed, by the tension of the belt **110**, and by a force generated by the mattress **10** which is being fed into the inner loop **115** and urges the inner loop **115** to expand. However, it is noted that while dedicated support structures may not be required, such support structures could nonetheless be present, e.g., in order to confine the inner loop **115** to a certain space or to avoid contact of the inner loop with other parts of the roll-packing machine **100**, **100**′.
[0044] The segments **111**, **112**, **113**, **114** of the belt **110** may be formed of a plastic material. In this way, excessive weight of the belt **110** may be avoided while at the same time ensuring sufficient rigidity of the segments **111**, **112**, **113**, **114**. On a side facing the mattress **10** The segments **111**, **112**,

113, **114** may be provided with a surface material and/or a surface structure (in FIG. **4**A indicated by cross-hatched regions) to provide a desired level of grip of the mattress **10** on the belt **110**. For example, the segments **111**, **112**, **113**, **114** could be provided with rubber elements which form or cover the cross-hatched regions. As illustrated, each of the segments **111**, **112**, **113**, **114** may be composed from multiple sub-segments **111**A, **111**B, **112**A, **112**B, **112**C joined together along the width direction W. However, a single-piece configuration of each of the segments **111**, **112**, **113**, **114** would be possible as well. In some scenarios, also the conveyor belt **131** and/or the conveyor belt **141** of the feed mechanism could be implemented as a segmented belt, e.g., with similar characteristics as explained in connection with FIGS. **4**A and **4**B.

[0045] FIGS. **5**A-**5**H further illustrate the roll-packing process. While FIGS. **5**A-**5**H assume a configuration of the roll-packing machine **100** as illustrated in FIG. **2**, it is noted that a similar process could also be implemented using a configuration with more additional loops of the belt **110**, e.g., using a configuration of the roll-packing machine **100**′ as illustrated in FIG. **3**.

[0046] FIG. **5**A illustrates an initial stage of the roll-packing process, in which the mattress **10** is inserted into the feed mechanism of the roll-packing machine **100**. In this initial stage of the roll-packing process, the inner loop **115** of the belt **110** is absent or has only a small size. As illustrated, the mattress is placed on the lower conveyor belt mechanism **130**, and the upper conveyor belt mechanism **140** is then lowered to sandwich the mattress **10** between the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140**. If desired, the mattress **10** may also be compressed between the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140**.

[0047] FIG. **5**B illustrates a next stage of the roll-packing process, in which the mattress **10** is sandwiched, and optionally compressed, between the lower conveyor belt mechanism **130** and the upper conveyor belt mechanism **140**. In this stage, the conveyor belt mechanisms **130**, **140** are operated to convey the mattress **10** toward the portion of the belt **110** between the entry rollers **121**, **122** until the mattress **10** eventually makes contact with the belt **110**.

[0048] FIG. **5**C illustrates a next stage of the roll-packing process, in which the mattress **10** has made contact with the belt **110**. In this stage, the conveyor belt mechanisms **130**, **140** are operated to further convey the mattress **10** toward the portion of the belt **110** between the entry rollers **121**, **122**. The mattress **10** exerts an inward force onto the belt **110**. At the same time the belt **110** is advanced. The advancing belt **110** grips the end of the mattress **10** and pulls it toward the entry roller **122**, thereby initiating rolling of the mattress **10**. By controlling the roller **121** to rotate at a higher rate than the roller **122**, the inner loop **115** is allowed to expanding under the force exerted by the mattress **10**, as illustrated in FIG. **5**D.

[0049] As illustrated by FIGS. **5**D to **5**F, the process of advancing the belt **110** while feeding the mattress **10** to the expanding inner loop **115** continues until the mattress **10** is fully received in the inner loop **115**. As illustrated in FIG. **5**F, this may result in the mattress **10** being rolled in multiple spiral-like windings. The additional belt length required for expansion of the inner loop **115** is provided by moving the roller **124** towards the roller **125**, thereby reducing the size of the additional loop formed by the rollers **123**, **124**, and **125**.

[0050] FIG. **5**G illustrates a next stage of the roll-packing process, in which the fully rolled mattress **10** is wrapped with the wrapping material to form the mattress roll **20** with the outer wrapping **25**. This is achieved by providing the wrapping material from the supply roll **161** into the inner loop **115** with the rolled mattress **10**, while continuing to advance the belt **110**. This causes rotation of the rolled mattress **10** in the inner loop **115**, with the wrapping material being pulled in between the belt **110** and the rolled mattress **10**. By completing one or more rotations of the rolled mattress **10** in the inner loop **115**, the rolled mattress **10** is wrapped with one or more layers of the wrapping material. The cutting/welding tool **162** may then be used to cut the wrapping material and weld or otherwise fixate the wrapping material around the rolled mattress **20**. At this point, the mattress roll is finished, but still enclosed in the inner loop **115**.

[0051] FIG. **5**H illustrates a next stage of the roll-packing process, in which the finished mattress roll **20** is ejected from the inner loop **115** and the roll-packing machine **100**. This is achieved by tilting the support structure **155**, thereby moving the entry roller **122** away from the entry roller **121**. The movement of the entry roller **122** opens the inner loop **115** so that the mattress roll **20** is released and can be removed. As illustrated in FIG. **5**H the movement of the entry roller **122** may cause that the portion of the belt **110** between the entry rollers assumes a straight-line form so that the mattress roll **20** can be removed from the roll-packing machine **100** by rolling along the belt **110**. This rolling of the mattress roll **20** can be facilitated by moving the entry roller **122** to a lower vertical position than the entry roller **121**, so that rolling of the mattress roll **20** from the roll-packing machine **100** can be driven or at least assisted by gravity force. Alternatively or in addition, removal of the mattress roll **20** from the roll-packing machine **100** could be accomplished by driving the belt **110** to convey the mattress roll **20** from the roll-packing machine **100**. For example, in the situation illustrated in FIG. **5**H, the belt **110** could be driven oppositely to the above-mentioned advancement direction A used during rolling of the mattress **10**. When moving the entry roller **122** away from the entry roller **121** a decrease of the length of the belt portion between the entry rollers **121**, **122** may be compensated by moving the roller **124** back away from the roller **125**.

[0052] FIG. **6** shows a flowchart for illustrating a method of roll-packing a mattress, e.g., the above-mentioned mattress **10**. The method of FIG. **6** may be used to implement the above-described concepts and may be performed by a roll-packing machine, e.g., the above roll-packing machine **100** or **100**′.

[0053] At block **610** the mattress is fed to a closed-loop belt of a belt drive mechanism, e.g., a belt drive mechanism as described in connection with FIG. **2** or a belt drive mechanism as described in connection with FIG. **3**. The belt may be a segmented belt, e.g., as described in connection with FIGS. **4**A and **4**B for the belt **110**. The feeding of the mattress may be performed by a feed mechanism, e.g., a feed mechanism based on one or more conveyor belt mechanisms as illustrated in FIGS. **2** and **3**. In some scenarios, the mattress may be compressed while being fed to the belt.

[0054] At block **620**, an inner loop of the belt is formed, e.g., like the above-mentioned inner loop **115**. An example of the formation of the inner loop is illustrated in FIGS. **5**C and **5**D, and at block **630**, the mattress is received in the inner loop, e.g., as illustrated by FIGS. **5**C to **5**E. As explained above, the belt drive mechanism may be provided with a first roller and a second roller between which the inner loop is formed, e.g., the above-mentioned entry rollers **121**, **122**. Formation of the inner loop may then involve individually controlling rotation of the first roller and the second roller. Specifically, by controlling the first roller and the second roller to rotate at different rates, the inner loop may be formed while the mattress fed to the belt exerts an inward force onto the belt.

[0055] At block **640** the mattress is rolled by advancing the belt while the mattress is received in the inner loop, e.g., as illustrated in FIGS. **5**C to **5**F. In some scenarios, this may involve adjusting the size of the inner loop while the mattress is being rolled. As explained above, the belt drive mechanism may be provided with a first roller and a second roller between which the inner loop is formed, e.g., the above-mentioned entry rollers **121**, **122**. Adjusting the size of the inner loop may then involve individually controlling rotation of the first roller and the second roller. Specifically, by controlling the first roller and the second roller to rotate at different rates, the size of the inner loop may be increased in accordance with an increase of the diameter of the rolled mattress.

[0056] At optional block **650**, the rolled mattress is wrapped. This may involve feeding a wrapping material into the inner loop with the rolled mattress and advancing the belt to rotate the rolled mattress in the inner loop and wrap the wrapping material around the rolled mattress. An example of such wrapping process is described in connection with FIG. **5**G.

[0057] At optional block **660**, the rolled mattress is ejected from the inner loop. As explained above, the belt drive mechanism may be provided with a first roller and a second roller between which the inner loop is formed, e.g., the above-mentioned entry rollers **121**, **122**. Ejecting the rolled

mattress may then involve displacing the first roller away from the second roller. An example of such ejection process is described in connection with FIG. **5**H.
[0058] It is noted that the flowchart of FIG. **6** is not intended to imply a strict order of the illustrated actions and that the illustrated actions may for example overlap or be otherwise combined with each other.
[0059] It is to be understood that the illustrated roll-packing machine **100**, **100′** and its operations are susceptible to various modifications, without departing from the illustrated concepts. For example, the geometry, number and arrangement of rolls in the belt-drive mechanism could be varied. Further, various types of feed mechanisms could be used for feeding the mattress to the belt and into the inner loop, e.g., feed mechanisms based on conveyor rollers or conveyor wheels. Still further, other way of fixating the mattress roll **20** could be used in addition or as an alternative to the outer wrapping **25**, e.g., fixation by welding a wrapping present on the mattress **10** itself or fixation by using one or more strings or bands around the rolled mattress **10**.

Claims

1. A roll-packing machine for packaging a mattress, the roll-packing machine comprising: a belt drive mechanism comprising a closed-loop belt; a feed mechanism configured to feed the mattress to the belt; a supply roll of wrapping material; a tool for cutting the wrapping material; wherein the belt drive mechanism is configured to form an inner loop of the belt, to receive the mattress in the inner loop, and to roll the mattress and one or more layers of the wrapping material by advancing the belt while the mattress and one or more layers of the wrapping material are received in the inner loop, wherein the belt drive mechanism comprises a first roller and a second roller, the inner loop being formed between the first roller and the second roller, wherein the belt drive mechanism is configured to adjust a size of the inner loop by individually controlling rotation of the first roller and the second roller, and wherein the first and second rollers may be separated, resulting in the inner loop being opened, so that a mattress roll formed and wrapped in the inner loop is released and removed.
2. The roll-packing machine of claim 1, wherein the first roller and the second roller are located at an entry into the inner loop.
3. The roll-packing machine of claim 1, wherein the belt drive mechanism is configured to increase the size of the inner loop while the mattress is being rolled.
4. The roll packing machine of claim 1, further comprising a belt tensioning mechanism configured to control tension of the belt while adjusting the size of the inner loop.
5. The roll packing machine of claim 4, wherein the belt tensioning mechanism is configured to control the tension of the belt by adjusting a size of one or more additional loops of the belt.
6. The roll packing machine of claim 5, wherein the belt drive mechanism comprises multiple rollers for guiding the belt, and wherein the tensioning mechanism is configured to adjust the size of the one or more additional loops by displacing two or more of the rollers relative to each other.
7. The roll-packing machine of claim 1, wherein only one of the first and second rollers is movable with a tiltable support structure.
8. The roll-packing machine of claim 7, wherein the roller movable with the tiltable support structure is below the other roller of the first and second rollers when the tiltable support structure is in a down position so gravity may assist moving the mattress roll along the belt and out of the roll-packing machine.
9. The roll-packing machine of claim 1, further comprising a tiltable support structure independent of the feed mechanism, wherein the first and second rollers may be separated by the tiltable support structure.
10. The roll-packing machine according to claim 1, wherein the belt is a segmented belt.
11. The roll-packing machine according to claim 1, wherein the feed mechanism is configured to

compress the mattress while feeding the mattress to the belt.

12. The roll-packing machine according to claim 1, wherein the feed mechanism comprises a lower conveyor belt mechanism and an upper conveyor belt mechanism above the lower belt mechanism, the upper and lower conveyor belt mechanisms being separate from the belt drive mechanism, the lower conveyor belt mechanism being stationary and the upper conveyor belt mechanism being movable from an upper position to a lowered position to sandwich the mattress between the lower and upper conveyor belt mechanisms when the upper conveyor belt mechanism is in its lowered position.

13. The roll-packing machine of claim 1, wherein the first and second rollers may be separated by tilting a tiltable support structure, resulting in the inner loop being opened, so that a mattress roll formed in the inner loop is released and removed by rolling along the belt above the tiltable support structure.

14. A roll-packing machine for packaging a mattress, the roll-packing machine comprising: a belt drive mechanism comprising a closed-loop belt; a feed mechanism configured to feed the mattress to the belt; a tiltable support structure independent of the feed mechanism, wherein the belt drive mechanism is configured to form an inner loop of the belt, to receive the mattress in the inner loop, and to roll the mattress by advancing the belt while the mattress is received in the inner loop, wherein the belt drive mechanism comprises a first roller and a second roller, the inner loop being formed between the first roller and the second roller, wherein the belt drive mechanism is configured to adjust a size of the inner loop by individually controlling rotation of the first roller and the second roller, and wherein the first and second rollers may be separated by tilting the tiltable support structure, resulting in the inner loop being opened, so that a mattress roll formed in the inner loop is released and removed and wherein the roll-packing machine is further configured to wrap the rolled mattress by: feeding a wrapping material into the inner loop with the rolled mattress; and advancing the belt to rotate the rolled mattress in the inner loop and wrap the wrapping material around the rolled mattress.

15. The roll-packing machine of claim 14, wherein the first roller and the second roller are located at an entry into the inner loop.

16. The roll-packing machine of claim 14, wherein the belt drive mechanism is configured to increase the size of the inner loop while the mattress is being rolled.

17. The roll packing machine of claim 14, further comprising a belt tensioning mechanism configured to control tension of the belt while adjusting the size of the inner loop.

18. The roll packing machine of claim 17, wherein the tensioning mechanism is configured to control the tension of the belt by adjusting a size of one or more additional loops of the belt.

19. The roll packing machine of claim 18, wherein the belt drive mechanism comprises multiple rollers for guiding the belt, and wherein the tensioning mechanism is configured to adjust the size of the one or more additional loops by displacing two or more of the rollers relative to each other.

20. The roll-packing machine of claim 14, wherein only one of the first and second rollers is movable with the tiltable support structure.

21. The roll-packing machine of claim 20, wherein the roller movable with the tiltable support structure is below the other roller of the first and second rollers when the tiltable support structure is in a down position so gravity may assist moving the mattress roll along the belt and out of the roll-packing machine.

22. The roll-packing machine according to claim 14, wherein the feed mechanism is configured to compress the mattress while feeding the mattress to the belt.

23. A roll-packing machine for packaging a mattress, the roll-packing machine comprising: a belt drive mechanism comprising a closed-loop belt; a feed mechanism configured to feed the mattress to the belt; a supply roll of wrapping material; a tool for cutting the wrapping material; wherein the belt drive mechanism is configured to form an inner loop of the belt, to receive the mattress in the inner loop, and to roll the mattress and one or more layers of the wrapping material by advancing the belt while the mattress and one or more layers of the wrapping material are received in the inner loop, wherein the belt drive mechanism comprises a first roller and a second roller, the inner loop being formed between the first roller and the second roller, wherein the belt drive mechanism is configured to adjust a size of the inner loop by individually controlling rotation of the first roller and the second roller, and wherein the first and second rollers may be separated, resulting in the inner loop being opened, so that a mattress roll formed and wrapped in the inner loop is released and removed; and wherein the roll-packing machine is further configured to wrap the rolled mattress by: feeding a wrapping material into the inner loop with the rolled mattress; and advancing the belt to rotate the rolled mattress in the inner loop and wrap the wrapping material around the rolled mattress.